and techniques involved?

A large variety of materials go into the making of a launch vehicle— both metal ic and non-metal ic, which include composites and ceramics. In metals, different types of stainless steel, al oys of aluminium, magnesium, titanium, copper, beryl ium, tungsten and molybdenum are used. Composite materials are composed of a mixture or combination of two or more constituents which differ in form and material composition and which are essential y insoluble in one another. The materials which combine may be metal ic, organic or inorganic. While other material combinations possible are virtual y unlimited, the most typical composites in launch vehicles are made of structural constituents, embedded in a matrix. We used a large variety of glass fibre reinforced plastic composites and opened avenues for the entry of Kevlar, polyamides and carbon-carbon composites. Ceramics are special types of baked clay used for microwave transparent enclosures. We considered using ceramics, but had to reject the idea then due to technological limitations.

Through mechanical engineering, these materials are transformed into hardware. In fact, of al the engineering